

ASSIGNMENT-2

Q1

1) First we represent each sequence as a matrix and multiply them in reverse order (since each new transformation is applied to the result of the previous one). We will use 4x4 homogeneous co-ordinates to capture both linear transformation and translation.

Step-1 Rotation through $-\pi/3$

$$R_y(\pi) = \begin{bmatrix} \cos(-\pi/3) & 0 & \sin(-\pi/3) \\ 0 & 1 & 0 \\ -\sin(-\pi/3) & 0 & \cos(-\pi/3) \end{bmatrix} = \begin{bmatrix} 1/2 & 0 & -\sqrt{3}/2 \\ 0 & 1 & 0 \\ \sqrt{3}/2 & 0 & 1/2 \end{bmatrix}$$

Step-2 Rotation of $\pi/4$

$$R_x(0) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos 0 & -\sin 0 \\ 0 & \sin 0 & \cos 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\pi/4) & -\sin(\pi/4) \\ 0 & \sin(\pi/4) & \cos(\pi/4) \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & \sqrt{2}/2 & -\sqrt{2}/2 \\ 0 & \sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix}$$

Combined Rotation is $R = R_x \cdot R_y$

$$R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \sqrt{3}/2 & -\sqrt{3}/2 \\ 0 & \sqrt{3}/2 & \sqrt{3}/2 \end{bmatrix} \begin{bmatrix} 1/2 & 0 & -\sqrt{3}/2 \\ 0 & 1 & 0 \\ \sqrt{3}/2 & 0 & 1/2 \end{bmatrix}$$

$$= \begin{bmatrix} 1/2 & 0 & -\sqrt{3}/2 \\ -\sqrt{6}/4 & \sqrt{3}/2 & -\sqrt{3}/4 \\ \sqrt{6}/4 & \sqrt{3}/2 & \sqrt{3}/4 \end{bmatrix}$$

Step-3 Reflection across the XZ Plane

Reflecting across the XZ-plane negates the Y-coordinate

$$S_{xz} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Then I multiply this with combined rotation to get the total linear transformation M:

$$M = S_{xz} R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1/2 & 0 & -\sqrt{3}/2 \\ -\sqrt{6}/4 & \sqrt{3}/2 & -\sqrt{3}/4 \\ \sqrt{6}/4 & \sqrt{3}/2 & \sqrt{3}/4 \end{bmatrix}$$

$$= \begin{bmatrix} 1/2 & 0 & -\sqrt{3}/2 \\ \sqrt{6}/4 & -\sqrt{2}/2 & \sqrt{3}/4 \\ -\sqrt{6}/4 & \sqrt{3}/2 & \sqrt{2}/4 \end{bmatrix}$$

Step-4

Translation

Finally I added translation vector $T = [2, 1, -3]$ into the 4th column to construct the overall

4x4 homogeneous transformation matrix T :

$$T = \begin{bmatrix} 1/2 & 0 & -\sqrt{3}/2 & 2 \\ \sqrt{6}/4 & -\sqrt{2}/2 & \sqrt{2}/4 & 1 \\ \sqrt{6}/4 & \sqrt{2}/2 & -\sqrt{2}/4 & -3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

2) To find the new co-ordinates, multiply the matrix T by the original homogeneous vector $V = [4, -2, 1, 1]^T$

$$V_{\text{new}} = T \begin{bmatrix} 4 \\ -2 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{2}(4) + 0(-2) - \frac{\sqrt{3}}{2}(1) + 2 \\ \frac{\sqrt{6}}{4}(4) - \frac{\sqrt{2}}{2}(-2) + \frac{\sqrt{2}}{4}(1) + 1 \\ \frac{\sqrt{6}}{4}(4) + \frac{\sqrt{2}}{2}(-2) + \frac{\sqrt{2}}{4}(1) - 3 \\ 1 \end{bmatrix}$$

Simplifying each row:

$$x' = 2 - \sqrt{3}/2 + 2 = 4 - \sqrt{3}/2$$

$$y' = \sqrt{6} + \sqrt{2} + \sqrt{2}/4 + 1 = 1 + \sqrt{6} + 5\sqrt{2}/4$$

$$z' = \sqrt{6} - \sqrt{2} + \sqrt{2}/4 - 3 = -3 + \sqrt{6} - 3\sqrt{2}/4$$

Mapped Origin:

The origin of the initial frame is $O = [0, 0, 0, 1]^T$. Multiplying T by the origin zeroes out the 3×3 linear transformation entirely, leaving only the translation vector.

The origin is mapped exactly to translation: $(2, 1, -3)^T$

3) We evaluate the combined rotation matrix R

$$R = \begin{bmatrix} 1/2 & 0 & -\sqrt{3}/2 \\ -\sqrt{3}/4 & \sqrt{3}/2 & -\sqrt{3}/4 \\ -\sqrt{3}/4 & \sqrt{3}/2 & \sqrt{3}/4 \end{bmatrix}$$

Angle of Rotation (θ):

$$\text{Tr}(R) = R_{11} + R_{22} + R_{33}$$

$$\text{Tr}(R) = \frac{1}{2} + \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} = \frac{2 + 3\sqrt{3}}{2}$$

$$\text{Tr}(R) = 1 + 2\cos\theta$$

$$\cos\theta = \frac{\text{Tr}(R) - 1}{2} = \frac{\frac{2 + 3\sqrt{3}}{2} - 1}{2} = \frac{3\sqrt{3} - 2}{4}$$

$$\theta = \cos^{-1}\left(\frac{3\sqrt{3} - 2}{4}\right)$$

Axis of Rotation (K):

The skew-symmetric part of R gives the axis vector scaled by $2\sin\theta$. We compute v_{skew} from $R - R^T$

$$R - R^T = \begin{bmatrix} 0 & \sqrt{3}/4 & \frac{2\sqrt{3} + \sqrt{3}}{4} \\ -\sqrt{3}/4 & 0 & -\frac{3\sqrt{3}}{4} \\ \frac{2\sqrt{3} + \sqrt{3}}{4} & \frac{3\sqrt{3}}{4} & 0 \end{bmatrix}$$

Extracting the vector components mapping to

$$V_{skew} = \begin{bmatrix} 0 & -2 & y \\ z & 0 & -x \\ y & x & 0 \end{bmatrix}$$

$$V_{skew} = \begin{bmatrix} 3\sqrt{2}/4 \\ -2\sqrt{3} + \sqrt{6} \\ -\sqrt{6}/4 \end{bmatrix}$$

To find the true unit axis K , we normalize V_{skew}

$$\text{The magnitude } \|V_{skew}\| = \sqrt{V_x^2 + V_y^2 + V_z^2}$$

$$= \sqrt{\frac{42 + 12\sqrt{2}}{4}}$$

Dividing by magnitude gives unit vector K :

$$K = \frac{1}{\sqrt{42 + 12\sqrt{2}}} \begin{bmatrix} 3\sqrt{2} \\ -2\sqrt{3} - \sqrt{6} \\ -\sqrt{6} \end{bmatrix}$$

4) Rodrigues formula states that $R = I + \sin\theta \cdot K + (1 - \cos\theta)K^2$, where K is skew symmetric matrix

$$\text{Un-normalized matrix } V = (R - R^T) = 2\sin\theta \cdot K$$

We can algebraically rewrite Rodrigues formula using V to make the verification clearer:

$$R = I + \frac{1}{2}V + \left(\frac{1 - \cos\theta}{4\sin^2\theta} \right) \cdot V^2$$

Using the identity $\sin^2 \theta = 1 - \cos^2 \theta = (1 - \cos \theta)(1 + \cos \theta)$, the formula reduces strictly to our known variables

$$R = 1 + \frac{1}{2}V + \frac{1}{4(1+\cos\theta)}V^2$$

Let's compute the leading diagonal term (R_{11}) to demonstrate the mathematical match:

• $V_{11} = 0$

$$\begin{aligned} \text{The first element of } V^2 &= -(v_y^2 + v_x^2) \\ &= -\left(\frac{12 + 6\sqrt{2} + 6 + 6}{16}\right) = -\frac{6 + 3\sqrt{2}}{4} \end{aligned}$$

$$\text{The scalar } \frac{1}{4(1+\cos\theta)} = \frac{1}{4\left(1 + \frac{3\sqrt{2}-2}{8}\right)} = \frac{2}{6+3\sqrt{2}}$$

Plugging these into the rearranged Rodrigues equation for R_{11} :

$$\begin{aligned} R_{11} &= 1 + \frac{1}{2}(0) + \left(\frac{2}{6+3\sqrt{2}}\right)\left(\frac{-6+3\sqrt{2}}{4}\right) \\ &= 1 - \frac{2}{4} = \frac{1}{2} \end{aligned}$$

This precisely $R_{11} = 1/2$ from our direct computation in step 1.

Evaluating the remaining components of $1 + \frac{1}{2}V + \frac{1}{4(1+\cos\theta)}V^2$ follows the identical algebraic correlation, yielding

$R_{22} = \sqrt{2}/2$, $R_{33} = \sqrt{2}/4$, and generating perfectly
 matched off-diagonal elements, proving that
 our age formulation derived is good.

Q3

Understanding the setup

The general pinhole camera projection model is given by

$$x = K [R|t] X$$

- x is the 2D image point in homogeneous co-ordinate (3×1)
- K is the intrinsic camera matrix (3×3)
- $[R|t]$ is the extrinsic matrix comprising rotation R (3×3) and translation t (3×1), which map co-ordinates from the world frame to the camera frame
- X is the 3D world point in homogeneous co-ordinate (4×1). We can express this as $X = [X_{3D}^T, 1]^T$, where X_{3D} is the cartesian 3D point (3×1).

Step-1: Define Camera 1 (C_1) Projection

The problem states that the first camera's frame of reference is used as the world coordinate frame.

Because the world frame and C_1 's frame are identical, there is no rotation or ~~translation~~ translation between them.

For C_1 , rotation is identity matrix I , and translation is a vector 0 .

$$x_1 = K_1 [I|0] \begin{bmatrix} X_{3D} \\ 1 \end{bmatrix}$$

Multiplying the extrinsic matrix by the 3D point extracts just the 3D co-ordinates

$$x_1 = K_1 X_{3D}$$

Step-2: Define Camera 2 (C_2) Projection

The Problem states that the second camera's orientation is obtained by a pure 3D rotation R applied to the first camera.

'Pure rotation' means the two camera's share the exact same optical center. Therefore, the translation vector between them is zero ($t=0$)

The extrinsic rotation matrix mapping the world (Camera 1) to Camera 2 is R

for camera 2

$$x_2 = K_2 [R|0] \begin{bmatrix} X_{3D} \\ 1 \end{bmatrix}$$

multiplying this out gives;

$$x_2 = K_2 R X_{3D}$$

Step-3

$$x_1 = K_1 X_{3D}$$

$$x_2 = K_2 R X_{3D}$$

find solve X_{3D} using camera 2's equation

We multiply both side from left by inverse of K_2 and R

$$K_2 R X_{30} = x_2$$

$$R^{-1} K_2^{-1} K_2 R X_{30} = R^{-1} K_2^{-1} x_2$$

$$X_{30} = R^{-1} K_2^{-1} x_2$$

Next, substitute this expression for X_{30} into Equation 1st equation

$$x_1 = K_1 (R^{-1} K_2^{-1} x_2)$$

grouping them together gives us the matrix H

$$x_1 = (K_1 R^{-1} K_2^{-1}) x_2$$

$$x_1 = H x_2$$

$$\text{matrix } H = K_1 R^{-1} K_2^{-1}$$

or

$$= K_1 R^T K_2^{-1}$$

(Since R is standard

orthogonal rotation matrix)

Step 4

To prove that H is an invertible 3×3 matrix

Dimensions: K_1 is a 3×3 , R^{-1} is 3×3 and K_2^{-1} is 3×3 . The product of three 3×3 matrices is a

3x3 matrix - Thus H , 3x3

Invertibility (Determinant $\neq 0$): A matrix product is invertible if and only if all the individual matrices are invertible.

Intrinsic matrices (K_1, K_2): An intrinsic camera matrix is an upper triangular matrix where the diagonal entries are the focal lengths (f_x, f_y) and 1.

Since real ~~camera~~ camera has non-zero focal lengths, the determinants (the product of the diagonal) is non-zero.

Therefore, K_1 and K_2 are invertible.

Rotation matrix (R): By definition, a valid 3D rotation matrix belongs to the special orthogonal group $SO(3)$. Its determinant is always exactly $+1$, meaning it is always invertible (and $R^{-1} = R^T$).

Because K_1 , R and K_2^{-1} are all independently invertible, their product H is also guaranteed to be invertible.

The inverse mapping from camera 1 to camera 2 would simply be $H^{-1} = K_2 R K_1^{-1}$.

Finally, the homography matrix: $H = K_2 R K_1^{-1}$

It is an invertible 3x3 matrix because it is the product of three invertible 3x3 matrices.

Computer Vision Assignment Report

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3. Camera Calibration

A. Dataset 1: Provided Dataset

1. Intrinsic Camera Parameters

- **Focal Length (f_x, f_y):** (956.27, 957.18)
- **Principal Point (c_x, c_y):** (368.82, 651.78)
- **Skew:** 0.0000

2. Extrinsic Camera Parameters (First 2 Images)

- **Image 1:**

- **Rotation Matrix (R):**

[-0.997024, -0.013935, -0.075816]

[0.025153, -0.988503, -0.149088]

[-0.072866, -0.150551, 0.985913]

- **Translation Vector (t):** [3.11495, 1.14316, 15.91389]

- **Image 2:**

- **Rotation Matrix (R):**

[0.998300, 0.053809, -0.022395]

[-0.050649, 0.991057, 0.123445]

[0.028837, -0.122101, 0.992098]

- **Translation Vector (t):** [-4.36650, -1.46641, 15.80877]

3. Radial Distortion Coefficients

- **Coefficients (k_1, k_2, p_1, p_2, k_3):** [0.19697, -0.70161, 0.00382, 0.00678, 0.09781]
- **Observation:** When looking at the raw images, the straight lines of the checkerboard (especially near the edges) appear slightly curved or bowed due to the physical shape of the camera lens. After applying the estimated distortion coefficients to undistort the images, these lines are mathematically remapped to their true geometric positions, causing the bowed edges to straighten out completely.

Original vs Undistorted (Provided Dataset)



Original



Undistorted



Original



Undistorted



Original



Undistorted



Original



Undistorted



4. Reprojection Error

- **Mean Error:** 0.0696 pixels
- **Standard Deviation:** 0.0230 pixels



5. Visualizing Reprojected Corners

- **How Reprojection Error is Computed:** The reprojection error is calculated by taking the known 3D real-world coordinates of the checkerboard (assuming $Z=0$ since it lies flat on a table) and mathematically projecting them back onto the 2D image plane using our newly estimated intrinsic matrix, extrinsic matrices, and distortion coefficients. The straight-line distance in pixels between where the algorithm actually found the corners versus where the math predicted they should be is the reprojection error.

image 25 green: detected, red: reprojected



image 24 green: detected, red: reprojected



image 23 green: detected, red: reprojected



image 22 green: detected, red: reprojected

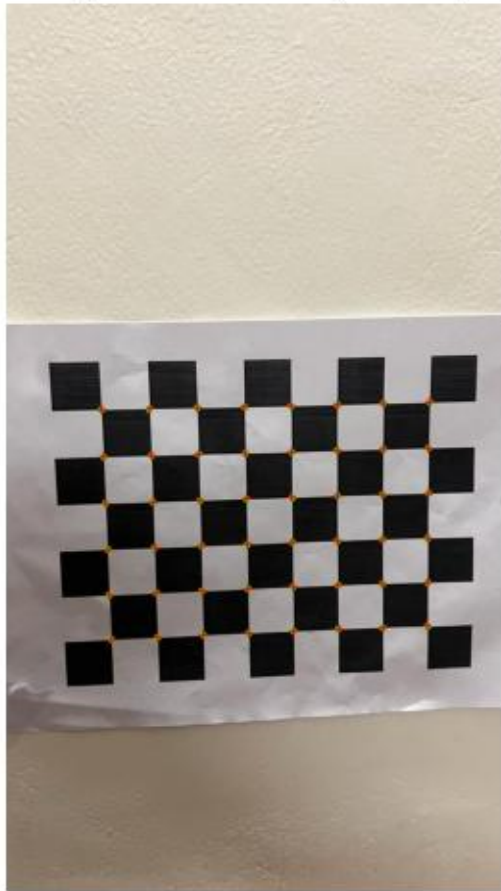


image 21 green: detected, red: reprojected

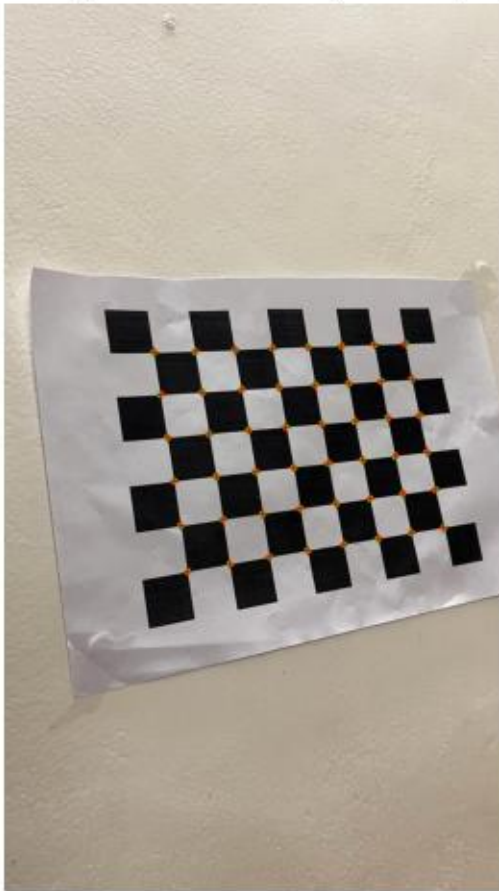


image 20 green: detected, red: reprojected

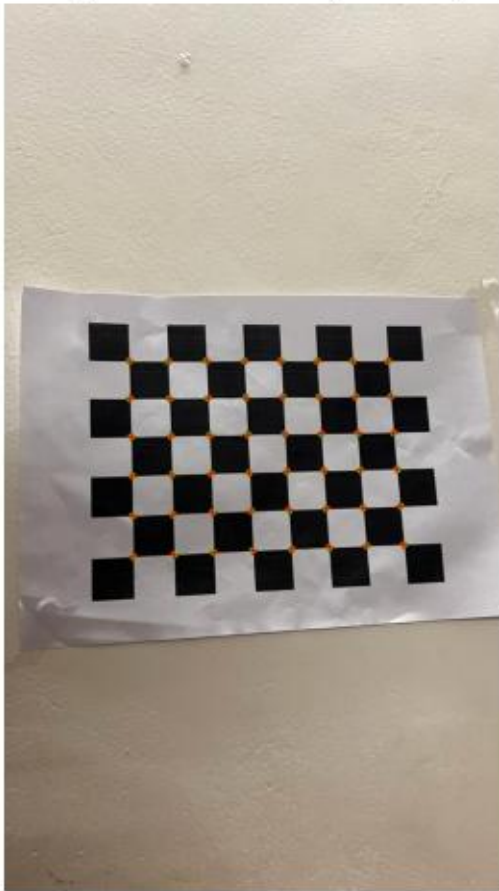


image 19 green: detected, red: reprojected

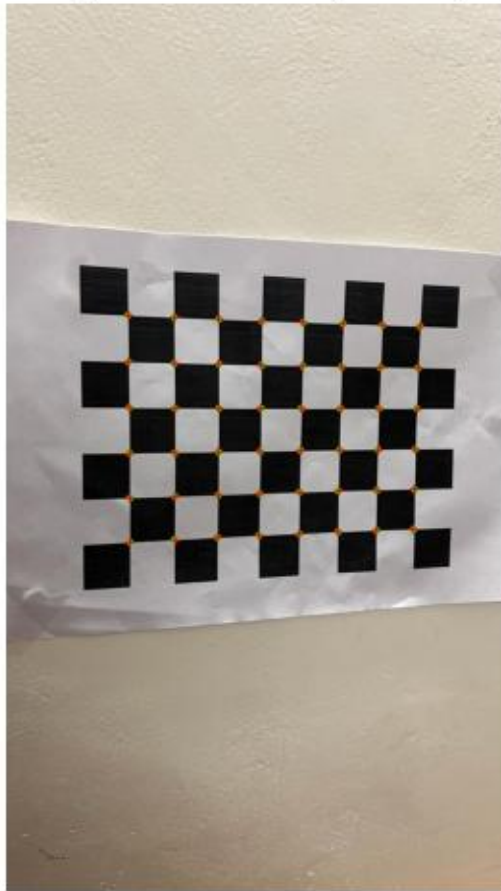


image 18 green: detected, red: reprojected



image 17 green: detected, red: reprojected

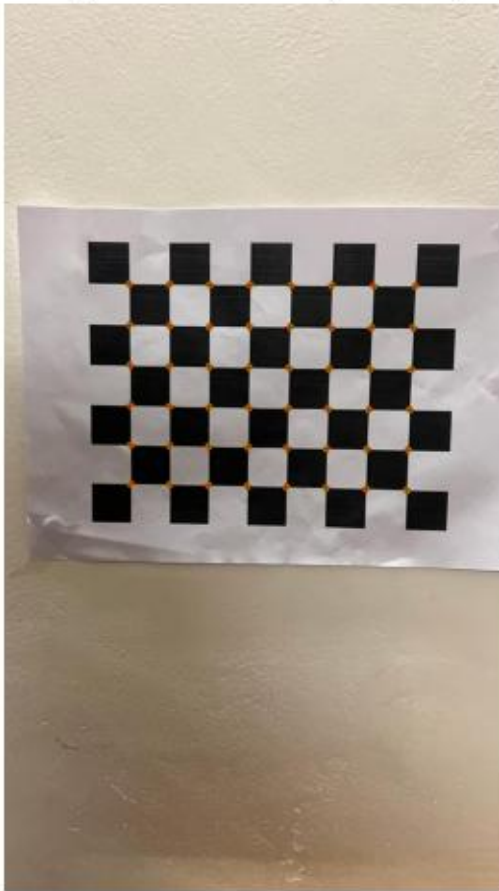


image 16 green: detected, red: reprojected

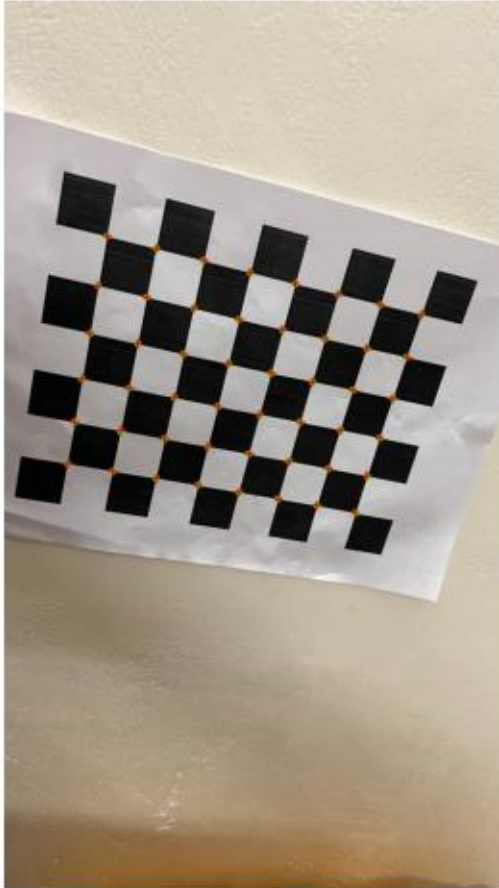


image 15 green: detected, red: reprojected

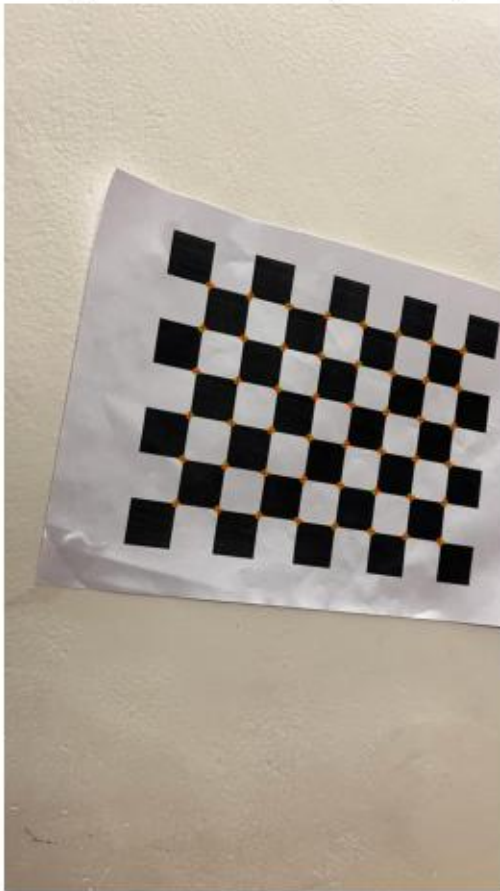


image 14 green: detected, red: reprojected

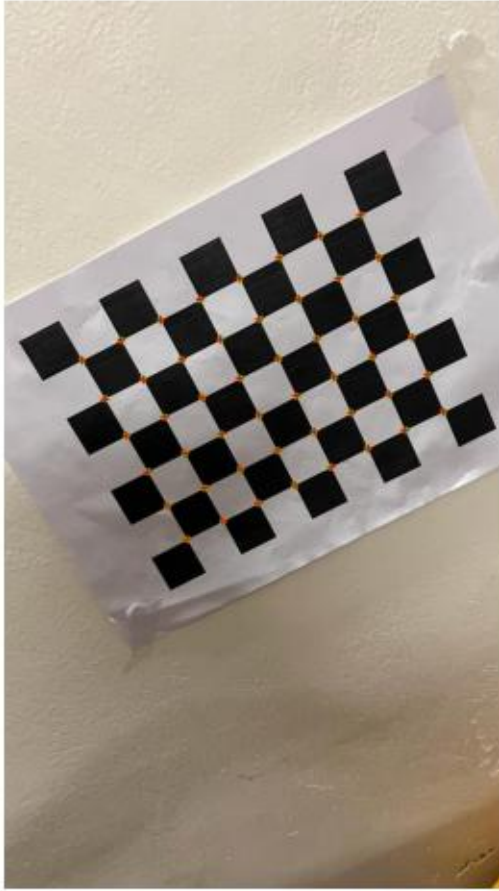


image 13 green: detected, red: reprojected

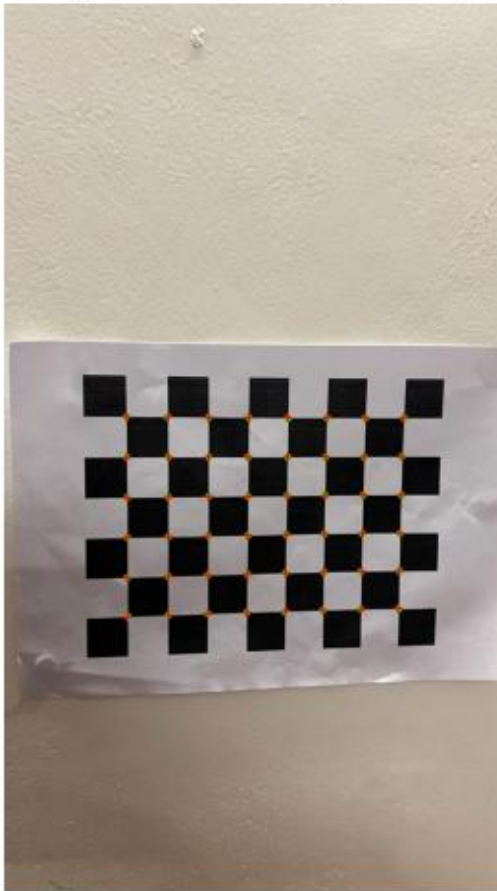


image 12 green: detected, red: reprojected

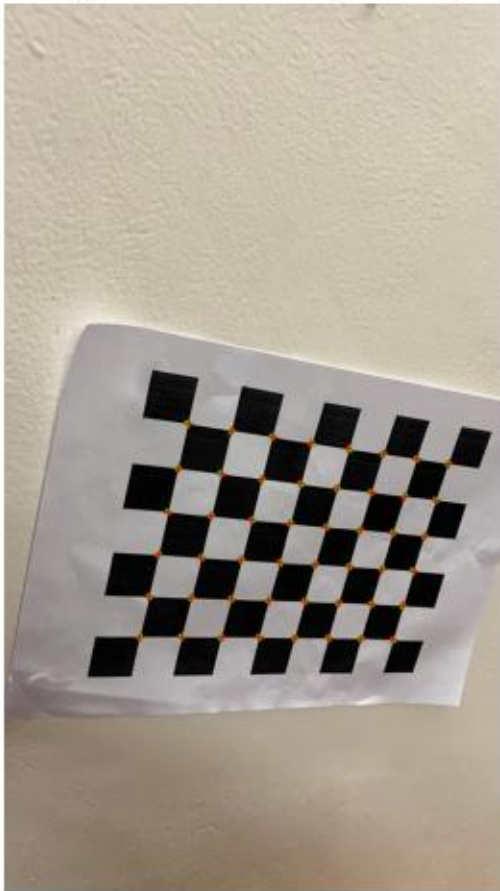


image 11 green: detected, red: reprojected

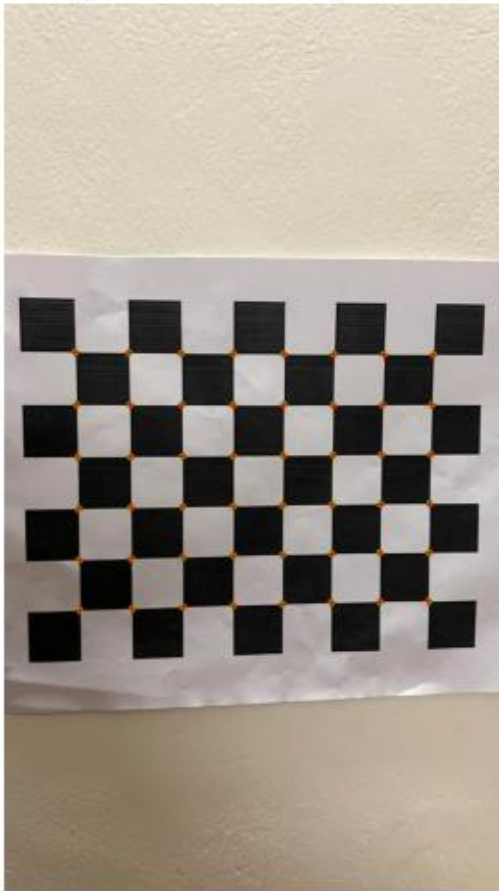


image 10 green: detected, red: reprojected

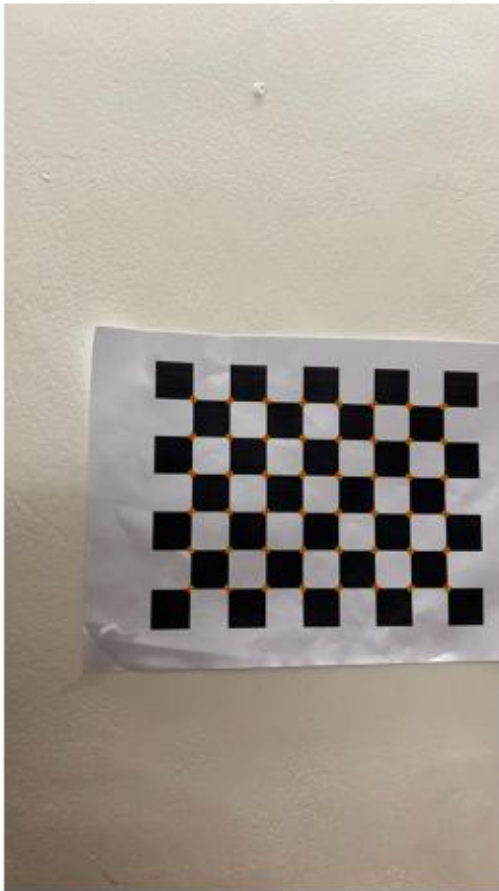


image 9 green: detected, red: reprojected

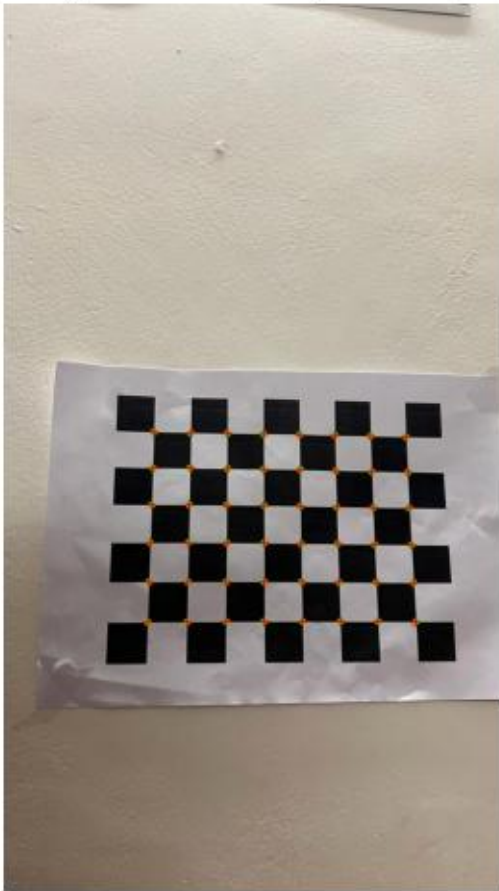


image 8 green: detected, red: reprojected

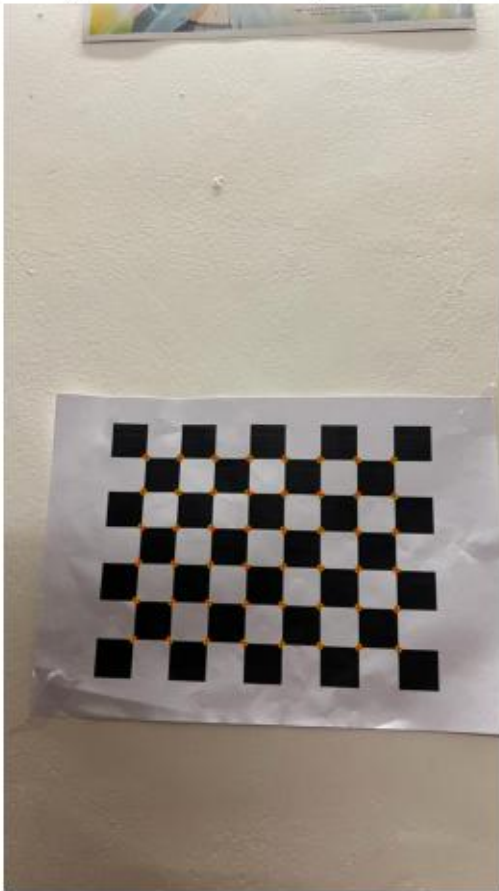


image 7 green: detected, red: reprojected

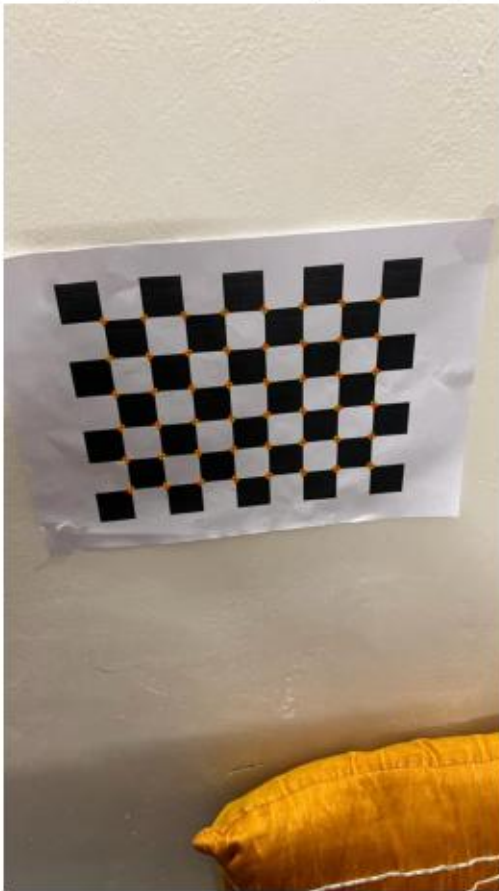


image 6 green: detected, red: reprojected



image 5 green: detected, red: reprojected

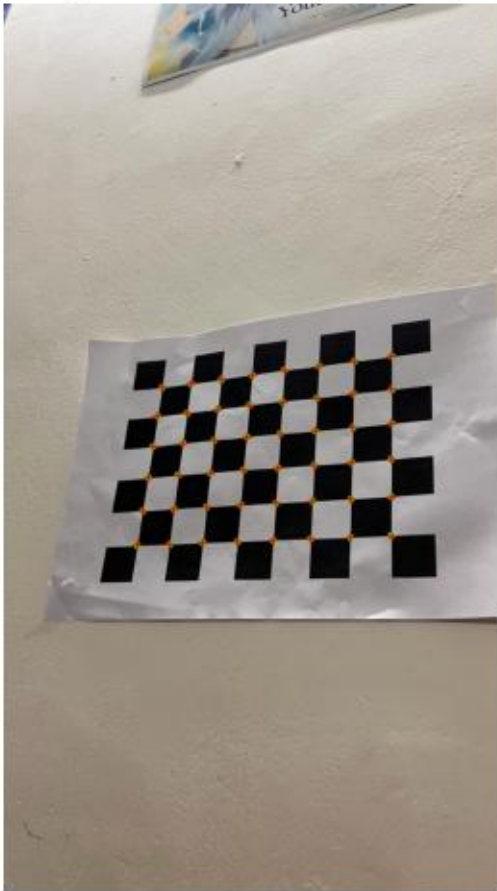


image 4 green: detected, red: reprojected



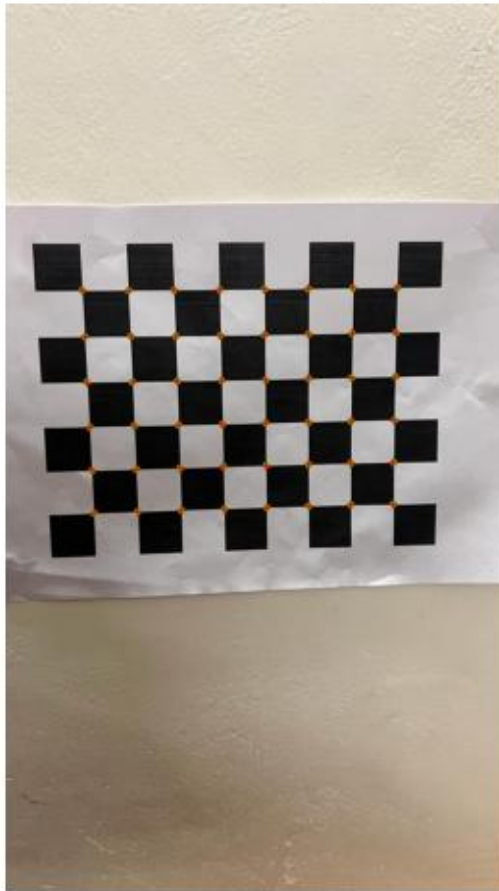
image 3 green: detected, red: reprojected



image 2 green: detected, red: reprojected



image 1 green: detected, red: reprojected



6. Checkerboard Plane Normals in Camera Frame

- Image 1: $[-0.07584, -0.14912, 0.98590]$
- Image 2: $[-0.02240, 0.12340, 0.99210]$
- Image 3: $[-0.39841, -0.18760, 0.89781]$
- Image 4: $[-0.34651, -0.52832, 0.77511]$
- Image 5: $[0.35825, 0.39821, 0.84443]$
- Image 6: $[0.04560, -0.32534, 0.94449]$
- Image 7: $[-0.12722, -0.50723, 0.85236]$
- Image 8: $[0.07148, 0.21655, 0.97364]$
- Image 9: $[-0.09455, 0.19676, 0.97588]$
- Image 10: $[-0.14539, 0.07289, 0.98668]$
- Image 11: $[-0.10742, -0.08763, 0.99034]$
- Image 12: $[-0.38489, -0.30961, 0.86947]$

- Image 13: [-0.13485, -0.10690, 0.98508]
- Image 14: [-0.09738, -0.27488, 0.95653]
- Image 15: [-0.36033, 0.01772, 0.93265]
- Image 16: [-0.16480, -0.36286, 0.91715]
- Image 17: [0.03637, -0.11185, 0.99305]
- Image 18: [-0.22287, 0.06766, 0.97249]
- Image 19: [-0.27088, -0.09693, 0.95771]
- Image 20: [-0.16571, 0.15925, 0.97323]
- Image 21: [-0.39744, 0.18990, 0.89775]
- Image 22: [-0.08290, -0.12807, 0.98829]
- Image 23: [-0.10295, -0.40211, 0.90978]
- Image 24: [0.18925, -0.06610, 0.97970]
- Image 25: [0.45424, 0.39937, 0.79634]

B. Dataset 2: Custom Dataset

1. Intrinsic Camera Parameters

- **Focal Length (fx, fy):** (1492.48, 1491.91)
- **Principal Point (cx, cy):** (678.31, 756.49)
- **Skew:** 0.0000

2. Extrinsic Camera Parameters (First 2 Images)

- **Image 1:**

- **Rotation Matrix (R):**

[-0.53371, 0.12894, 0.83577]

[-0.42352, -0.89618, -0.13218]

[0.73197, -0.42451, 0.53291]

- **Translation Vector (t):** [-0.96929, 2.58713, 11.22663]

- **Image 2:**

- **Rotation Matrix (R):**

[0.98155, 0.17449, 0.07815]

[-0.18843, 0.81364, 0.54997]

[0.03237, -0.55455, 0.83151]

- **Translation Vector (t):** [-2.49609, -4.83198, 15.28216]

3. Radial Distortion Coefficients

- **Coefficients (k1, k2, p1, p2, k3):** [-0.01552, -0.48608, -0.00623, -0.01003, 0.85205]

Original vs Undistorted (Custom_Dataset)

Original



Undistorted



Original



Undistorted



Original



Undistorted



Original



Undistorted



Original



Undistorted



4. Reprojection Error

- **Mean Error:** 0.4399 pixels
- **Standard Deviation:** 0.0693 pixels



5. Visualizing Reprojected Corners

image 25 green: detected, red: reprojected

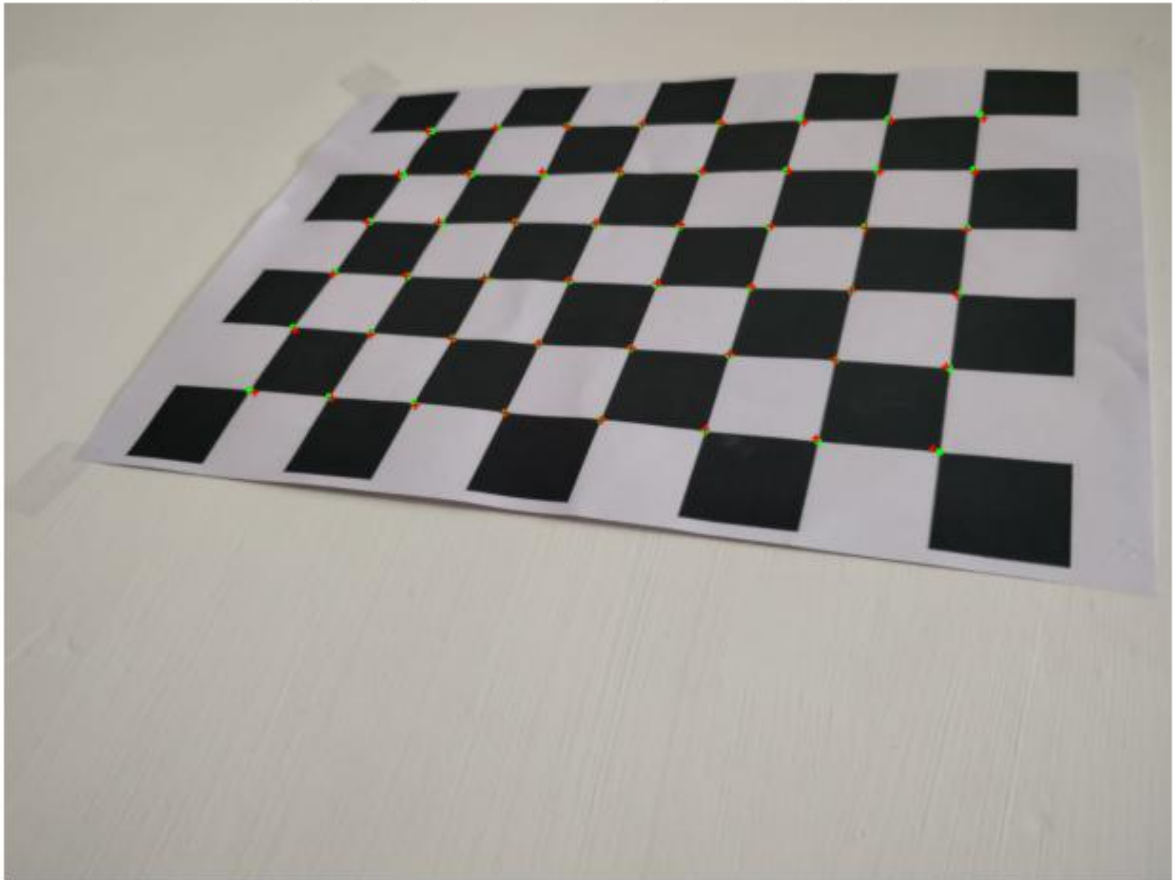


image 24 green: detected, red: reprojected



image 23 green: detected, red: reprojected



image 22 green: detected, red: reprojected

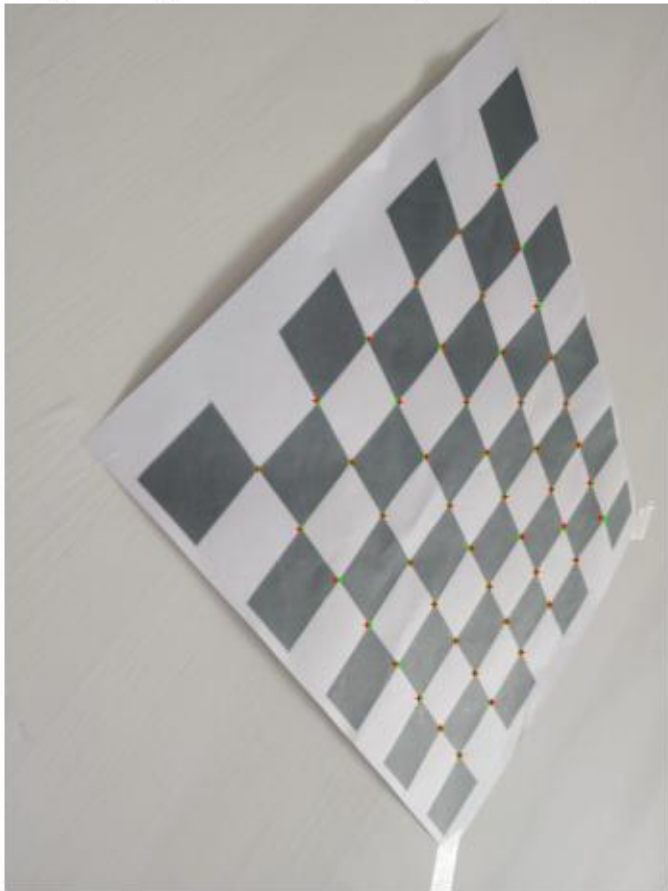


image 21 green: detected, red: reprojected



image 20 green: detected, red: reprojected

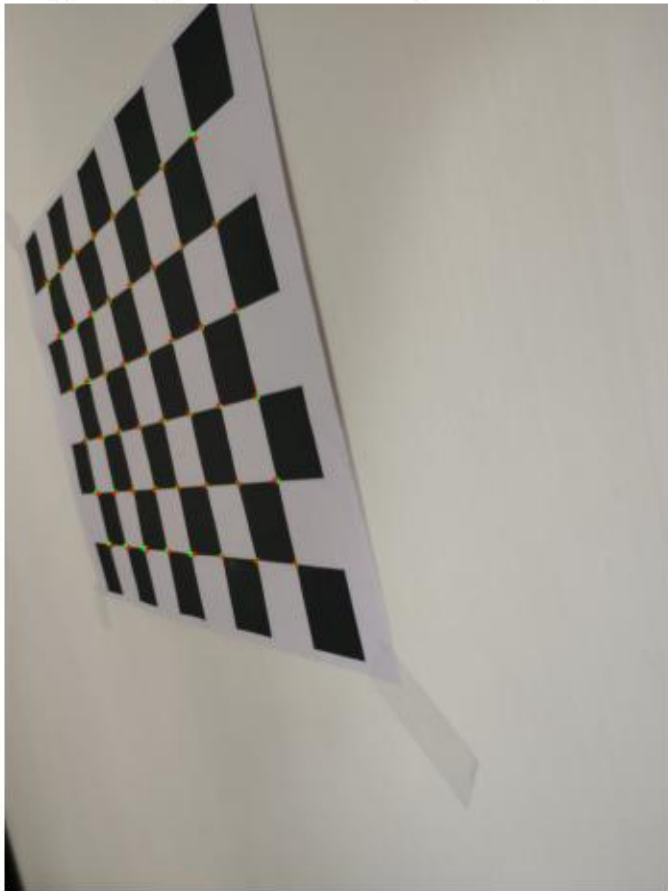


image 19 green: detected, red: reprojected

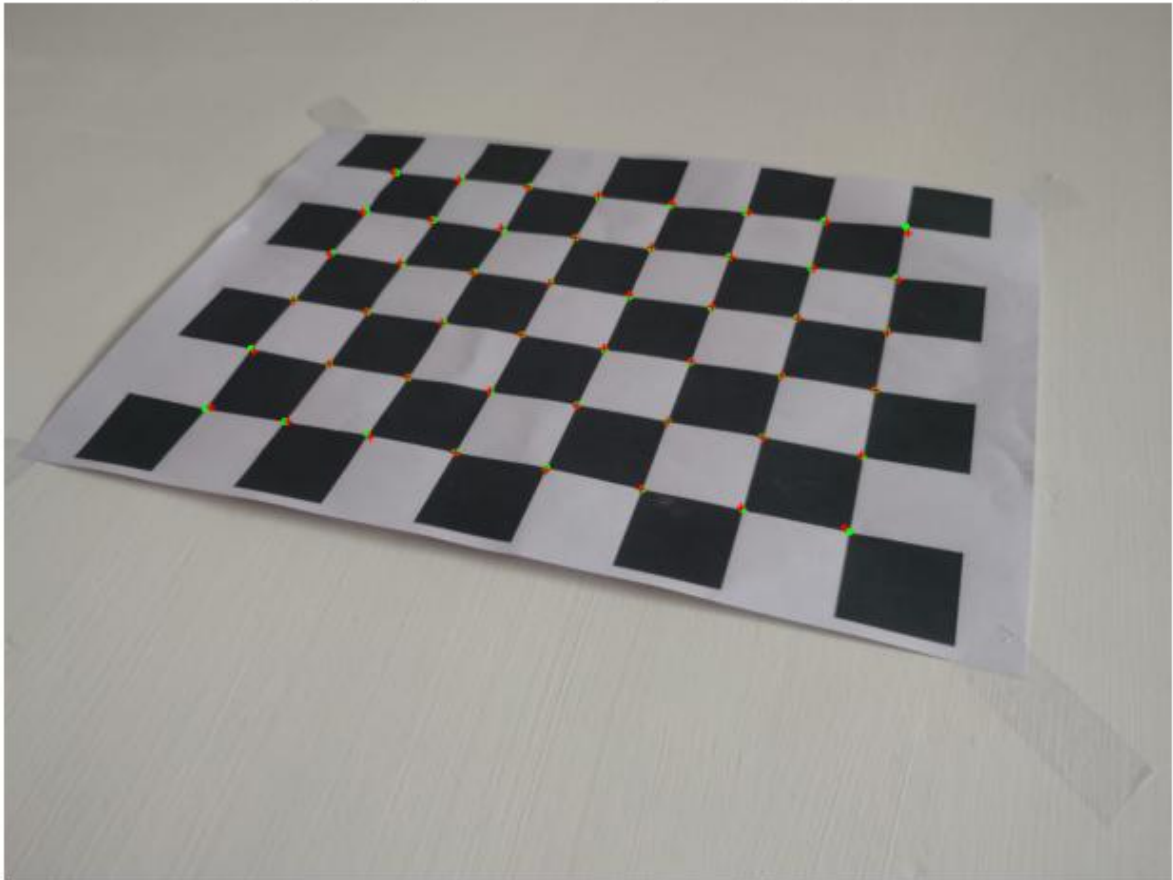


image 18 green: detected, red: reprojected



image 17 green: detected, red: reprojected



image 16 green: detected, red: reprojected



image 15 green: detected, red: reprojected



image 14 green: detected, red: reprojected



image 13 green: detected, red: reprojected



image 12 green: detected, red: reprojected



image 11 green: detected, red: reprojected



image 10 green: detected, red: reprojected



image 9 green: detected, red: reprojected



image 8 green: detected, red: reprojected



image 7 green: detected, red: reprojected



image 6 green: detected, red: reprojected



image 5 green: detected, red: reprojected



image 4 green: detected, red: reprojected



image 3 green: detected, red: reprojected

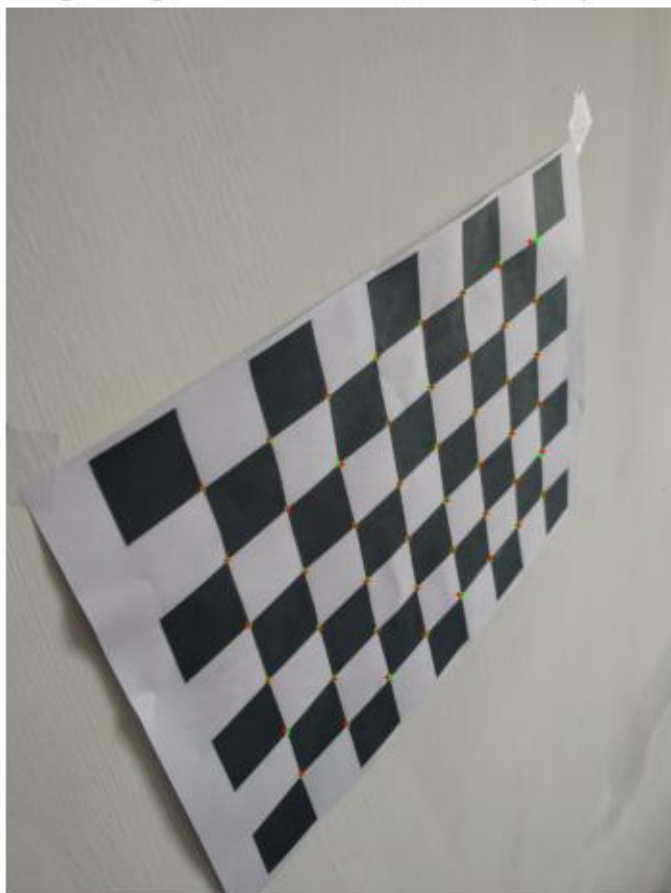
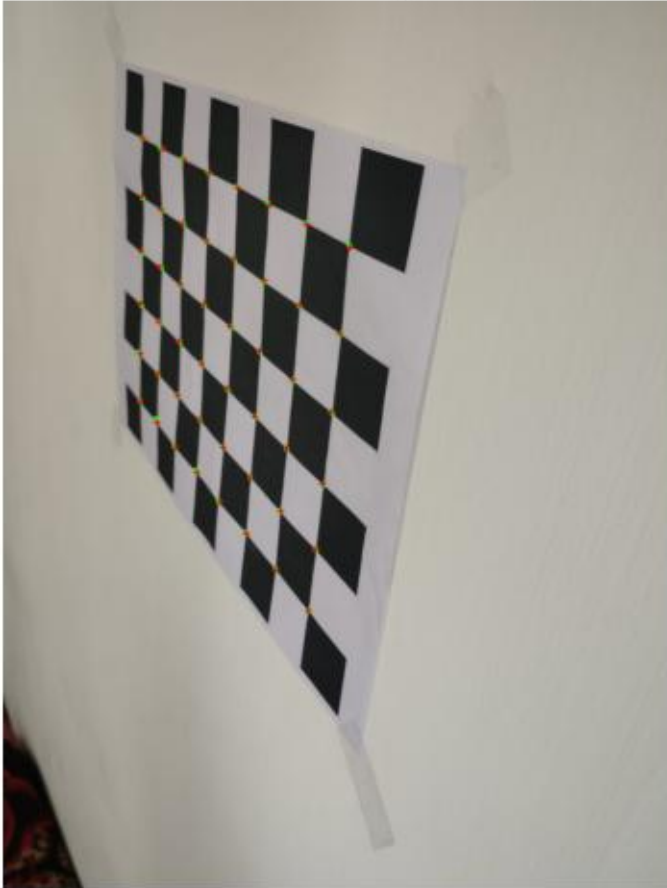


image 2 green: detected, red: reprojected



image 1 green: detected, red: reprojected



6. Checkerboard Plane Normals in Camera Frame

- Image 1: [0.83577, -0.13218, 0.53291]
- Image 2: [0.07815, 0.54997, 0.83151]
- Image 3: [-0.78972, -0.22378, 0.57118]
- Image 4: [0.09200, -0.87831, 0.46914]
- Image 5: [-0.01947, 0.67649, 0.73618]
- Image 6: [0.09891, -0.79814, 0.59428]
- Image 7: [-0.01553, -0.91198, 0.40992]
- Image 8: [0.01718, -0.24659, 0.96896]
- Image 9: [0.79058, 0.15948, 0.59122]
- Image 10: [-0.09203, -0.79386, 0.60108]
- Image 11: [-0.83599, -0.10684, 0.53822]
- Image 12: [0.75202, -0.18215, 0.63346]
- Image 13: [-0.81374, -0.16332, 0.55780]

- Image 14: [0.35341, -0.78593, 0.50735]
- Image 15: [0.79513, -0.10399, 0.59745]
- Image 16: [-0.82658, -0.11825, 0.55025]
- Image 17: [0.81714, -0.09572, 0.56843]
- Image 18: [-0.85803, -0.02788, 0.51284]
- Image 19: [0.11316, 0.64862, 0.75264]
- Image 20: [0.79083, -0.20432, 0.57691]
- Image 21: [-0.79641, -0.09835, 0.59669]
- Image 22: [-0.75191, -0.19207, 0.63065]
- Image 23: [-0.53687, -0.28538, 0.79392]
- Image 24: [-0.83890, -0.11464, 0.53206]
- Image 25: [0.25368, 0.61420, 0.74725]

4. Panorama Generation (40 points)

1. Keypoint Detection (SIFT)

- Keypoints and feature descriptors were successfully extracted from the first two images using the SIFT algorithm.

Image 1 SIFT Keypoints

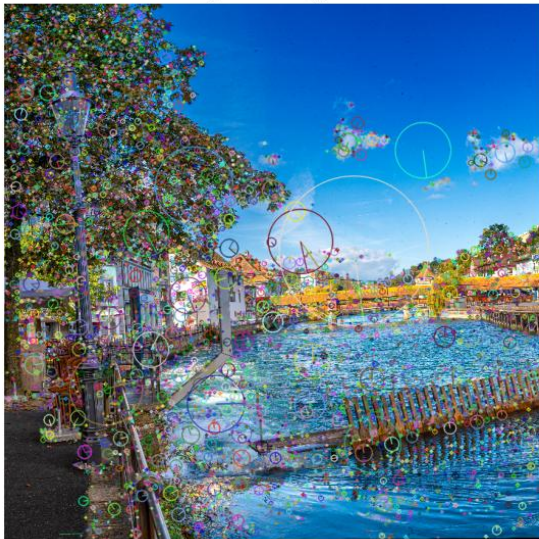
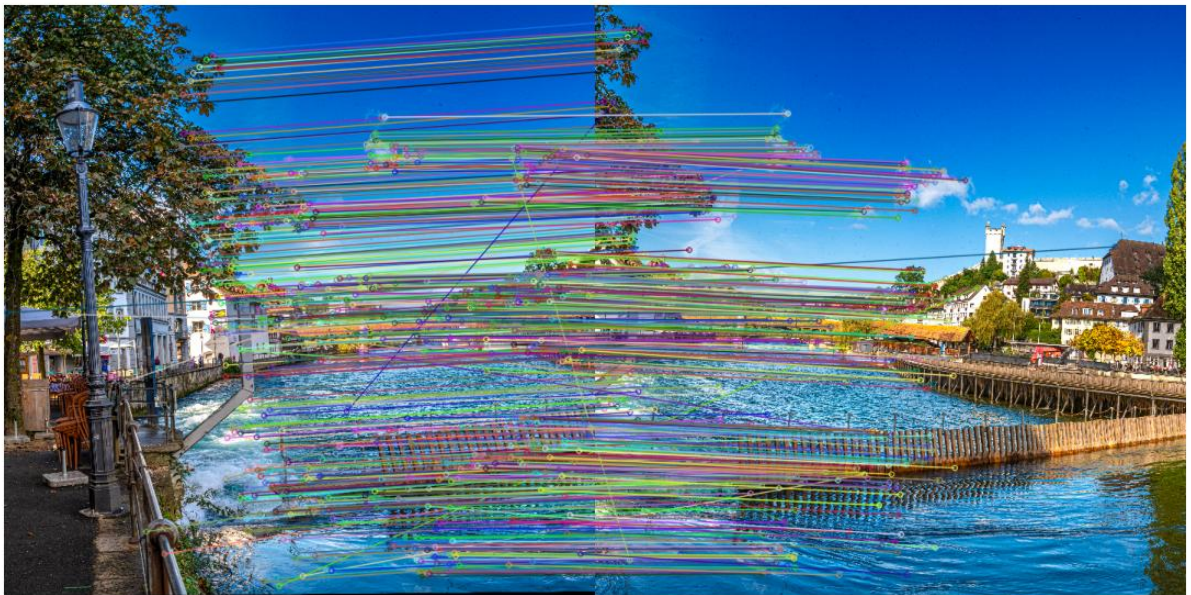


Image 2 SIFT Keypoints



2. Feature Matching

- Extracted features were matched using both the Brute Force algorithm and the optimized FLANN based matcher.



3. Homography Estimation

- The Homography matrix between the images was estimated utilizing RANSAC to ignore bad matches and outliers. The computed matrix is:

[1.089213, 0.144209, 189.18438]

[-0.157417, 0.988461, 42.73470]

[0.00000004, -0.00000039, 1.00000]

4. Perspective Warping

- Image 2 was successfully warped into the perspective plane of Image 1 using the estimated Homography matrix above.

Original Image 1



Warped Image 2



5. Stitching

Unblended / Uncropped Panorama



Cropped Panorama



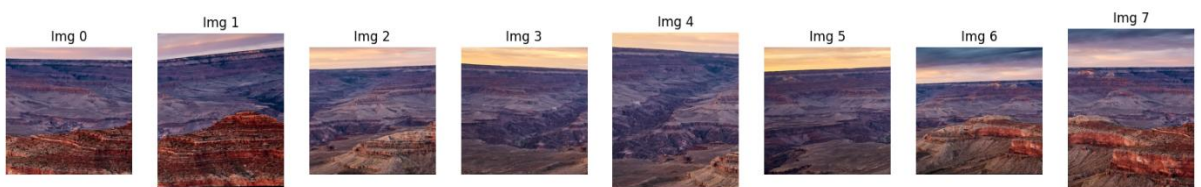
6. Multi-Stitching

- K-means clustering successfully separated the mixed dataset into 3 distinct scene sets based on color histograms. The automated multi-stitching script was then run on each cluster.

Cluster 0 Images (Order of Stitching)



Cluster 1 Images (Order of Stitching)



Cluster 2 Images (Order of Stitching)



pano 0



pano 1



pano 2



References

- [OpenCV Documentation: Camera Calibration and 3D Reconstruction](#)
- [OpenCV Documentation: Feature Matching with FLANN](#)
- [SciPy Documentation: Clustering with K-Means Algorithm](#)